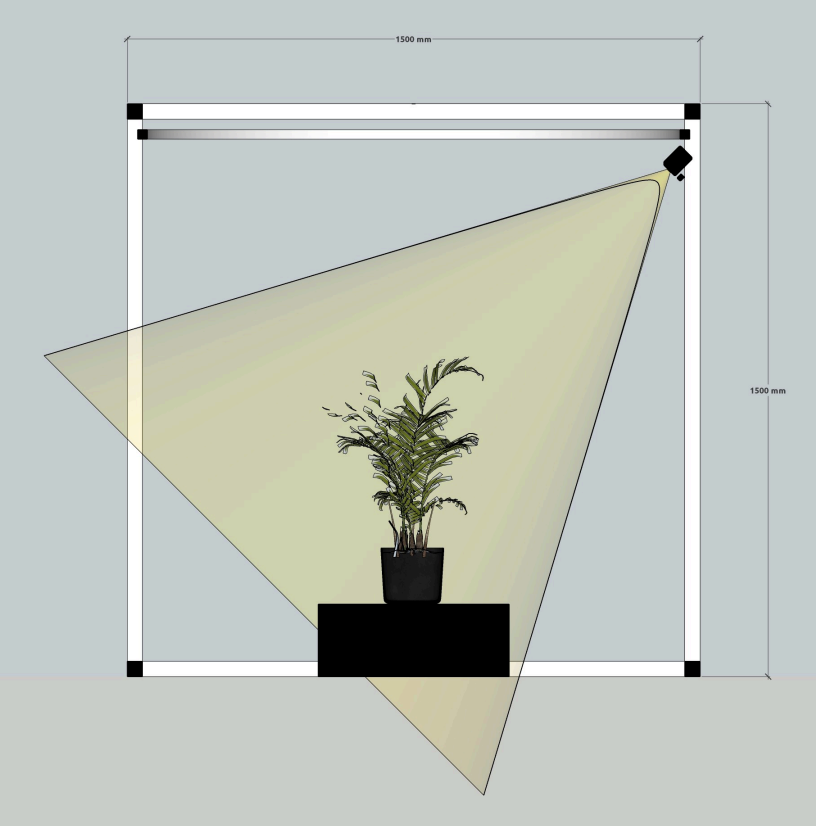


1. Multi-view image acquisition

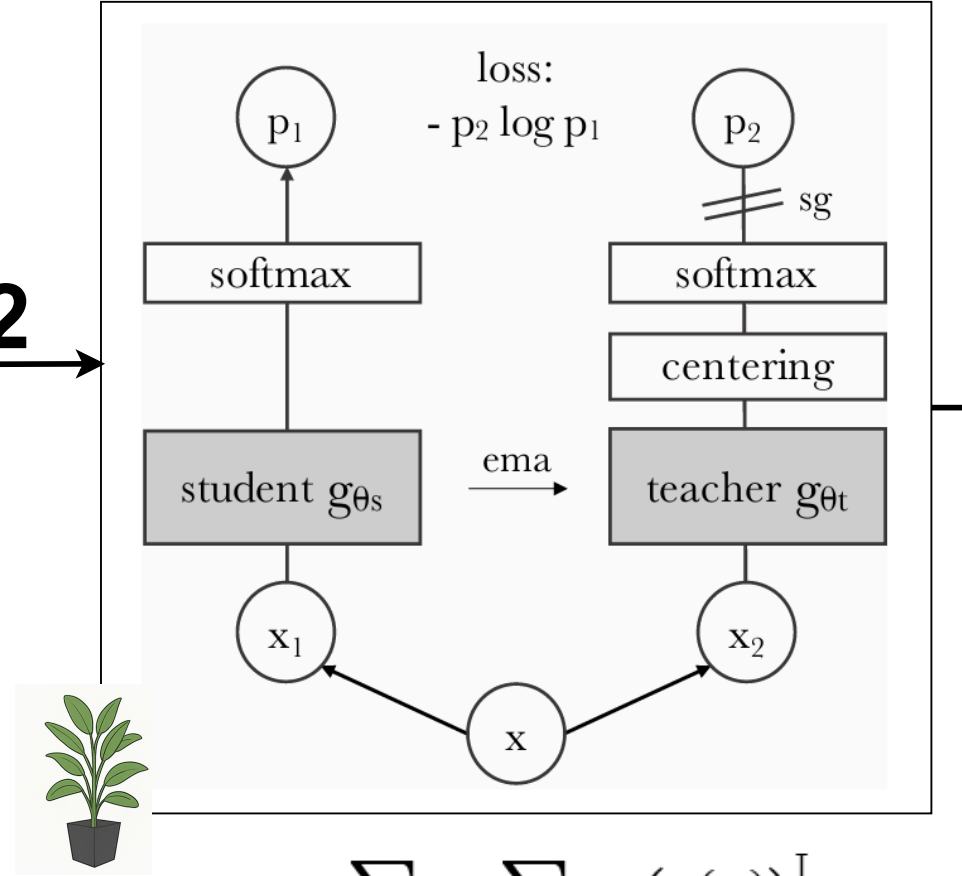


$$D = \{(I_v, K_v, E_v)\}_{v=1}^V$$

$$E_v = [\mathbf{F}_v | t_v] \in SE(3)$$

$$K_v \in R^{3 \times 3}$$

3. DINO Vision Transformer Backbone



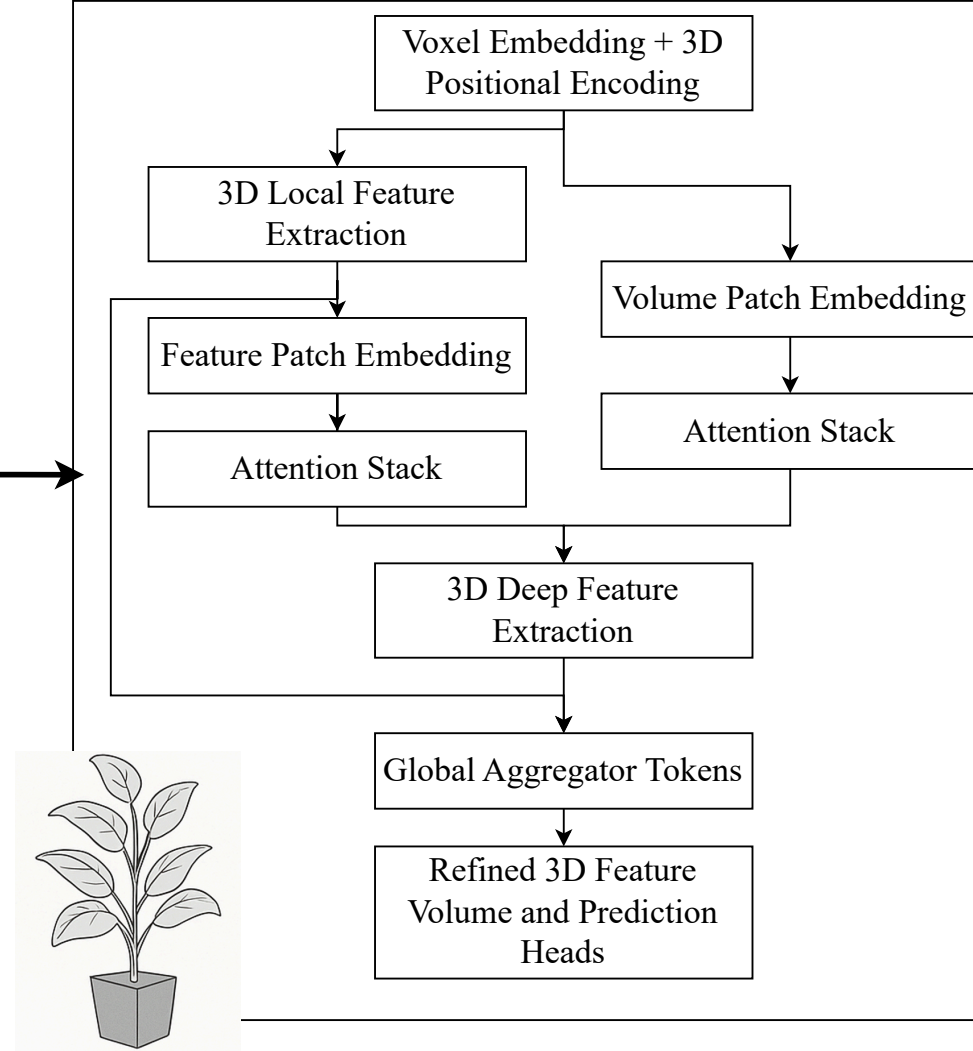
$$\mathcal{L}_{DINO} = - \sum_{c_s \in \mathcal{C}_v} \sum_{c_t \in \mathcal{C}_v^{global}} (P_t^{(c_t)})^\top \log P_s^{c_s}$$

$$F_v(u) = \text{Interp}(T_s^{full}, u) \in \mathbb{R}^d$$

4

INFER
Initial 3D Feature Volume

5. Volumetric Transformer (3D reasoning)

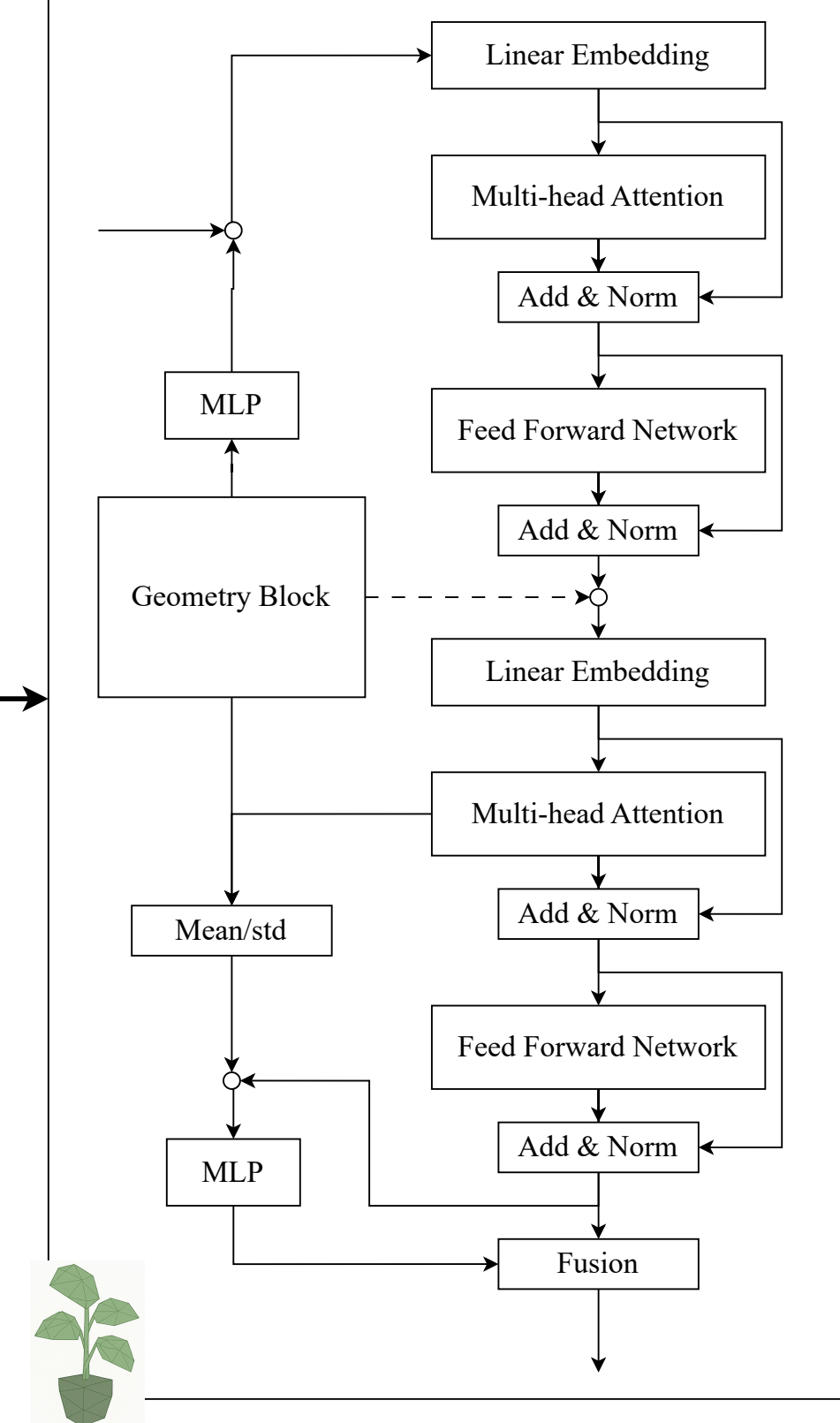


$$Q = H^{(l)} W_Q, K = H^{(l)} W_K, V = H^{(l)} W_V$$

$$\alpha_{ij} = \text{softmax} \left(\frac{Q_i K_j^\top}{\sqrt{d}} + b(x_i - x_j) \right) \quad \hat{h}_i = \sum_j \alpha_{ij} V_j$$

$$h_i^{(l+1)} = h_i^{(l)} + \text{MLP} \left(\text{LN}(\hat{h}_i) \right)$$

6. Visual Geometry Grounding

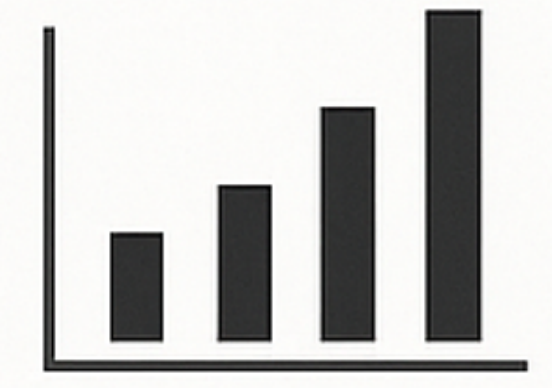


$$\beta_{iv} = \text{softmax}_v \left(\frac{q_i k_{iv}^\top}{\sqrt{d_g}} \right)$$

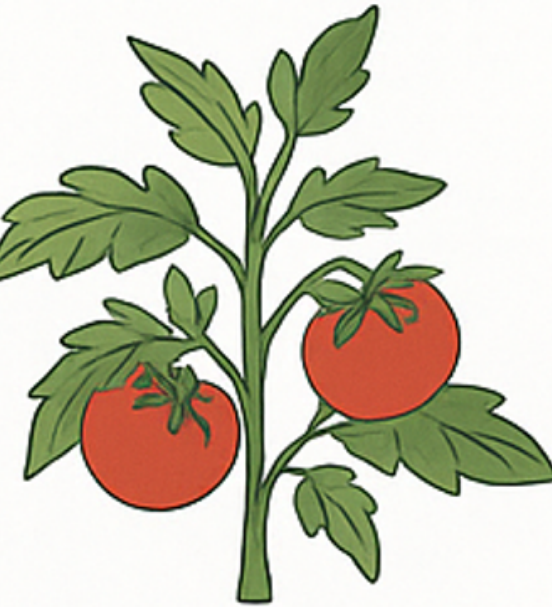
$$\tilde{h}_i = h_i^{(L)} + \sum_v \beta_{iv} v_{iv}$$

$$k_{iv} = W_K^g \bar{f}_v(x_i), v_{iv} = W_K^g \bar{f}_v(x_i), q_i = W_K^g h_i^{(L)}$$

7



Biomass Estimation



Species Recognition